

Nayem Sarker

929-350-2258 | nayemsarker321@gmail.com | LinkedIn | GitHub

EDUCATION

Queens College, City University of New York
Bachelor of Arts in Computer Science

Queens, NY
December 2025

EXPERIENCE

SpaceQ

January 2025 – August 2025

Senior AI Quantum Engineer

New York, NY

- Designed a vector-based AI memory engine, enabling semantic storage and NLP retrieval within an R&D platform.
- Integrated Python logging, SHA-256 hashing, Qdrant vector search, and FastAPI for a scalable network framework.
- Validated system with routine cosine similarity testing, achieving 92% match accuracy across 700+ text embeddings.
- Refined engine reasoning with PBFT consensus algorithms, reducing decision latency by 33% in decentralized governance simulations.

Google

July 2024 – Present

G-SWEP Fellow

New York, NY

- Enhanced technical skills by solving 15+ programming challenges weekly in direct collaboration with Google SWEs.

Xtrades

February 2022 – March 2024

AI and Data Engineer

New York, NY

- Led a team of 6 for the launch of an AI-driven stock/options trading analytics app used by 315,000+ users.
- Developed a Python ML-trade algorithm, integrating Scikit-learn and XGBoost models into Kafka pipelines to deliver 250+ monthly automated trades with 72% accuracy.
- Migrated user and trade metadata into PostgreSQL architecture, enabling efficient aggregation for analytics.
- Optimized marketing outreach through automation with Python schedulers, boosting MoM user growth by 12%.
- Built a cross-platform chat by integrating Discord APIs, REST endpoints, and in-app messaging via WebSockets.

CT (BloFin Brokerage)

May 2022 – January 2024

Software Engineer

New York, NY

- Integrated copytrading programs to Discord using Python and Flask, reducing user trade execution time by 65%.
- Scaled system architecture with Redis to cache trader sessions and copy signals, enabling support for 50,000+ users.
- Maintained 99% uptime by hosting services on AWS and orchestrating microservice management with Kubernetes.
- Deployed a full-stack e-commerce checkout system with React frontend, Node.js backend, and AWS database.

Deloitte

May 2021 – July 2021

Apprentice

New York, NY

- Collaborated with Deloitte SWEs in researching deepfake detection strategies across video and text media.
- Built a prototype detection system in PyTorch using discriminator logic, achieving 70% accuracy in early testing.
- Led a team of 4 to present research weekly to executives, refining communication skills in an Agile environment.

PROJECTS

MMAJudgerAI | *Python, Scikit-learn, YOLOv8, PyTorch, MediaPipe, OpenCV*

- World's first AI model that analyzes MMA footage and creates scores based on action (kicks, punches, takedowns).
- Trained Pytorch LSTM networks with 40+ hours of footage to detect skeletal keypoints, achieving a 96% F1-score.
- Enhanced judging accuracy by 50% through YOLOv8 system integration, enabling the evaluation of action impact.

Stock Trader Ranking Leaderboard | *Python, Kafka, Airflow, PostgreSQL, REST APIs, Pandas, React.js*

- Recognized in a UX Product of the Month award and highlighted by Nasdaq for innovation.
- Processed 10,000+ equity trades to compute volatility, win rate, and risk metrics among 2,800+ active trader profiles.

SKILLS

Languages: Python, SQL, Java, JavaScript, TypeScript, HTML/CSS, C#, C++

Frameworks/Libraries: PyTorch, Hugging Face, LangChain, Node.js, React, Redis, Kubernetes, Kafka, FastAPI, Flask

Tools/Platforms: AWS (EC2, RDS, SageMaker), Azure, Docker, Git, PostgreSQL